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CS604 TODAY 7:45 AM

27-JAN-25

MCQS;

1: ---------- contain the name of the other files and/or directories

Ans= **Directory**

2: ---------- occurs in a dynamic memory allocation system when many of the free block are too small to satisfy any request.

Ans= **Fragmentation**

3: consider a process A is trying to access a page that does not belong to its address space which of the following protection scheme again such action under paging?

Ans= **valid/invalid (v) bit**

4: in ---------- scheme, number of frames allocated to a process is proportional to its size.

Ans= **proportional allocation**

5: When a process needs to execute, memory is requested by the process. The size of the process is compared with the amount of adjacent memory available to execute the process. This method is called _____.

Ans= Contiguous Allocation

6: In case of thrashing if CPU utilization is too low, the operating system ---------- the degree of multiprogramming.

Ans= **increase**

7: assume a logical address space 16 page of word 1024 words, each mapped into physical memory of 32 frames each word consist of 2 bytes what will be the total number of bits required for f (frames)?

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Ans= **5**

8: with -------- you can used condition variable.

Ans= **Monitors**

9: MMU stand for

Ans= **Memory management unit**

10: --------- is a separation of user logical memory from physical memory.

Ans= **Virtual Memory**

11: External fragmentation take place in

Ans= **Multiprogramming with variables task**

12: in the scan algorithm, disk ---- start at one end of disk and moved toward the other end.

Ans= **Arm**

13: You can use ---------- to synchronize cooperative process.

Ans= **Semaphores**

14: A dashed line is used to represent a --------- in resource allocation graph.

Ans= **Claim Edge**

15: The integer value of -------- semaphores can range over an unrestricted integer domain.

Ans= **Counting**

16: ------- is used in the detection and recovery mechanism to handle deadlock.

Ans= **Wait for graph**

17: The critical section problem can be solved by the following except

Ans= **Firmware based solution**

18: the most important property of the working set is its

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Ans= **size**

19: Which command, Display permissions and some other attributes for prog1.c in your current directory?

Ans**= ls –l prog1.c**

20: ----------keep in memory only those instructions and data that are needed at any given time.

Ans= **Overlays**

21: If the system can allocate resources to each process in some order and still avoid a deadlock then it is said to be in _____ state.

Ans= **Safe**

22: A condition where a set of blocked processes each holding a resource and waiting to acquire a resource held by another process in the set is termed as _____.

Ans= **Deadlock** **(This MCQ appeared twice in the exam)**

23: In pages segmentation, the logical address is legal if d is _____segment length.

Ans= **< (less then)**

24: A_____ is an integer variable that, apart from initialization is accessible only through two standard atomic operations: wait and signal.

Ans= **Semaphore**

25: In the Bakery Algorithm:

Ans= **lowest number is served next.**

Q26: -------- is a way to establish connection btw the file to be shared.

Ans= **Link**

I don't remember the other MCQs that appeared in the exam.

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Short Question:

Q1: Scenario given

Related to scenario define data structure of Baker algorithm.

Ans=

Baker's algorithm utilizes several key data structures to manage resource allocation and prevent deadlocks. The **Available[]** array holds the number of available instances for each resource type. The **Allocation[][]** matrix tracks the resources allocated to each process, while the **Request[][]** matrix stores the resources requested by each process. The **Max[][]** matrix represents the maximum resources each process may need. The **Work[]** array temporarily holds the available resources during execution, and the **Finish[]** array keeps track of whether a process has completed its execution. These structures together ensure safe resource management and avoid deadlock situations.

- Q2: What are the issues that can be caused due to incorrect use of semaphores? You are required to write 3 issues.

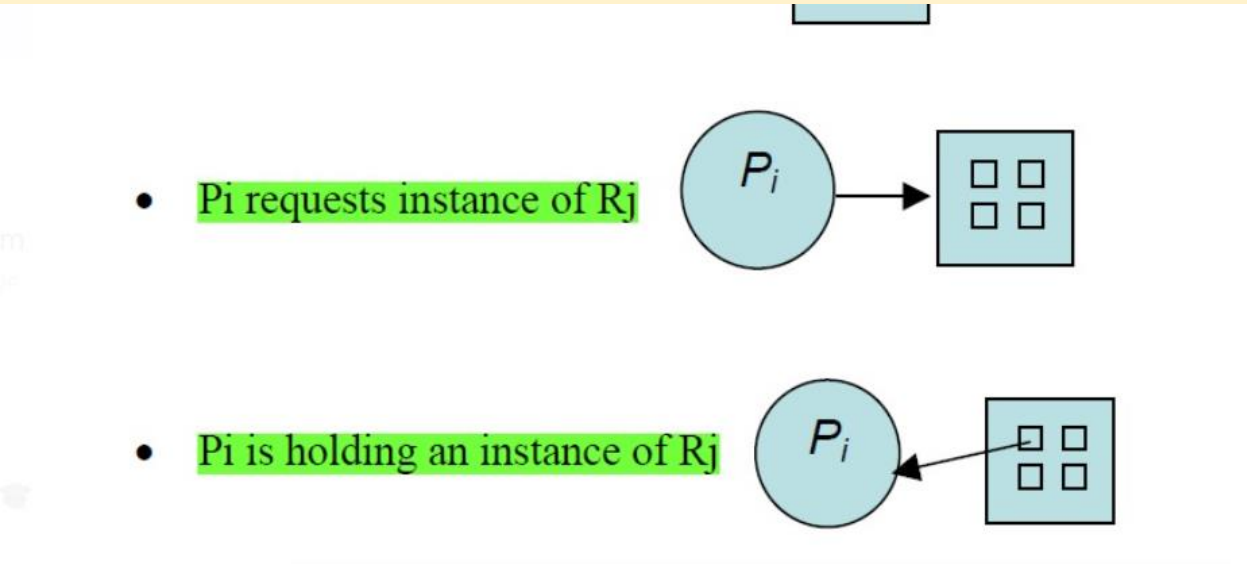
Ans=

- Livelock
  - Livelock occurs when two or more processes are actively trying to acquire a semaphore.
- Deadlock
  - A deadlock occurs when two or more processes are blocked indefinitely
  - Starvation
- Starvation
  - occurs when a process is unable to gain access to a semaphore

Q3: Diagram given req mention names.

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Ans=



- Q4: Consider a swapping system in which memory consists of the following hole sizes in memory order: 10KB, 4KB, 20KB, 18KB, 7KB, 9KB, 12KB and 15KB. Which holes is taken for successive segment requests of: 12KB, 10KB, 9KB for the first fit?

Ans= In a swapping system, the first fit algorithm allocates memory to the first available hole that is large enough to accommodate the requested segment. Here's the step-by-step allocation for the given segment requests:

Memory Holes in Order:

- 10KB
- 4KB
- 20KB
- 18KB
- 7KB
- 9KB
- 12KB
- 15KB

Segment Requests:

1. **12KB**
2. **10KB**
3. **9KB**

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First Fit Allocation:

1. **12KB request:**
    - The first hole that can fit 12KB is the 20KB hole.
    - After allocating 12KB, the remaining hole is 8KB.
    - Memory holes after allocation: 10KB, 4KB, 8KB (remaining from 20KB), 18KB, 7KB, 9KB, 12KB, 15KB.
2. **10KB request:**
    - The first hole that can fit 10KB is the 10KB hole.
    - After allocation, the 10KB hole is completely used up.
    - Memory holes after allocation: 4KB, 8KB (remaining from 20KB), 18KB, 7KB, 9KB, 12KB, 15KB.
3. **9KB request:**
    - The first hole that can fit 9KB is the 9KB hole.
    - After allocation, the 9KB hole is completely used up.
    - Memory holes after allocation: 4KB, 8KB (remaining from 20KB), 18KB, 7KB, 12KB, 15KB.

Summary of Allocations:

- **12KB request**: Allocated from the 20KB hole (leaving 8KB).
- **10KB request**: Allocated from the 10KB hole.
- **9KB request**: Allocated from the 9KB hole.

Q5: SCENRIO GIVEN ABOUT PAGE FALUT (COMPLETE QUESTION NOT REM)

Q6: SCENRIO GIVEN ACCORDING TO SCERRIO CONTIGOUS ALLOCTAION

Long questions:

Q1 numerical:

Q2: About page fault (why page fault occurs who's responsible for that and solution).

Q3. What is semaphores. Also define classical definitions

of wait and signal with the help of code.

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Ans=

Semaphores *Important*

Hardware solutions to synchronization problems are not easy to generalize to more complex problems. To overcome this difficulty we can use a synchronization tool called a semaphore. A **semaphore** S is an integer variable that, apart from initialization is accessible only through two standard atomic operations: wait and signal. These operations were originally termed P (for wait) and V (for signal). The classical definitions of wait and signal are:

decrement

```
wait(S) {
   while(S<=0)
   ;// no op
   S--;
}
```

wait reduce value

increment

```
signal(S) {
   S++;
}
```

signal increase value

Q4: About protection (Complete question not rem)

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